# B2 Algebra and quadratics: problems

← Module overview · Solutions

Give exact answers (fractions and square roots) unless a question asks for a decimal. Show your working: in a school test (*toets*) most marks are for the method.

## Practice

**1.** Expand and simplify (*haakjes wegwerken*).

(a) $3x(2x-5)$ (b) $(x+4)(x-7)$ (c) $(2x-3)^2$ (d) $(5a+2b)(5a-2b)$ (e) $(x+1)(x^2-x+1)$

**2.** Use a special product (*merkwaardig product*) to compute without a calculator.

(a) $51^2$ (b) $49^2$ (c) $103 \cdot 97$ (d) $999^2$

**3.** Factorise by taking out a common factor (*buiten haakjes brengen*).

(a) $6x^2 - 9x$ (b) $4a^2b - 10ab^2$ (c) $x(x-3) + 5(x-3)$

**4.** Factorise with the product-sum method.

(a) $x^2 + 7x + 12$ (b) $x^2 - 2x - 15$ (c) $x^2 - 11x + 24$ (d) $x^2 + x - 56$

**5.** Factorise completely.

(a) $x^2 - 81$ (b) $4x^2 - 25y^2$ (c) $3x^2 - 48$ (d) $x^4 - 16$

**6.** Factorise.

(a) $2x^2 + 7x + 3$ (b) $6x^2 - x - 2$ (c) $3x^2 - 10x + 8$

**7.** Solve with the product-is-zero rule (*product-is-nul*).

(a) $(x-4)(2x+3) = 0$ (b) $x^2 = 7x$ (c) $x^2 - 5x - 24 = 0$ (d) $4x^2 - 9 = 0$

**8.** Solve.

(a) $x^2 = 49$ (b) $(x-3)^2 = 10$ (c) $2(x+1)^2 - 18 = 0$ (d) $x^2 + 5 = 1$

**9.** Completing the square (*kwadraat afsplitsen*).

(a) Write $x^2 + 6x - 4$ in the form $(x+p)^2 + q$.
(b) Solve $x^2 - 8x + 5 = 0$ by completing the square.
(c) Write $2x^2 + 12x + 7$ in the form $a(x+p)^2 + q$.

**10.** Use the discriminant (*discriminant*) to find the number of solutions, then solve with the abc-formula (*abc-formule*) where possible.

(a) $2x^2 - 3x - 4 = 0$ (b) $x^2 + 4x + 4 = 0$ (c) $3x^2 + x + 2 = 0$ (d) $x^2 - 2x - 1 = 0$

**11.** Given the parabola $y = -2x^2 + 8x + 10$.

(a) Is it a *dalparabool* or a *bergparabool*? Explain.
(b) Find the vertex (*top*) and the axis of symmetry (*symmetrieas*).
(c) Find the points of intersection with both axes.
(d) Sketch the parabola.
(e) Find the points of intersection with the line $y = 2x + 10$.

**12.** Solve these rational equations (*gebroken vergelijkingen*). Check your answers.

(a) $\dfrac{2x+1}{x-3} = 3$ (b) $\dfrac{6}{x} = x - 1$ (c) $\dfrac{x^2}{x-3} = \dfrac{9}{x-3}$ (d) $\dfrac{x^2-4}{x+1} = \dfrac{x^2-4}{2x-3}$

**13.** Solve the inequalities using a sketch.

(a) $x^2 - x - 6 < 0$ (b) $2x^2 \geq 5x + 3$ (c) $x^2 + 2x + 5 > 0$

## School-exam level

**14. Thrown ball.** Lotte throws a ball upwards from the roof of a building. The height of the ball in metres after $t$ seconds is

$$h = -5t^2 + 20t + 25$$

(a) How high is the roof?
(b) Calculate the maximum height of the ball and the time at which it is reached.
(c) After how many seconds does the ball hit the ground?
(d) For how many seconds is the ball higher than $40$ m?

**15. Fence against a wall.** A farmer has $60$ m of fencing. He makes a rectangular enclosure against a long straight wall, so he only needs fence on three sides. Let $x$ be the length in metres of each side perpendicular to the wall.

(a) Show that the area is $A = x(60 - 2x)$.
(b) Calculate the maximum area and the dimensions that give it.
(c) For which values of $x$ is the area exactly $400$ m$^2$?

**16. Path around a lawn.** A rectangular lawn of $12$ m by $8$ m is surrounded by a path of constant width $x$ m. The lawn and path together cover $140$ m$^2$. Calculate the width of the path.

**17. Odd numbers.** The product of two consecutive odd numbers is $255$. Find all possible pairs.

**18. Parameters.**

(a) For which values of $p$ does $x^2 + px + 9 = 0$ have exactly one solution?
(b) For which values of $p$ does this equation have no solutions?
(c) For which value of $p$ does the line $y = x + p$ touch (*raken*) the parabola $y = x^2 - 3x + 5$? Give the point where they touch.

**19. Cinema tickets.** A cinema sells $300$ tickets per evening at €10. Research shows that for every €0.50 price increase, $10$ fewer tickets are sold. Let $n$ be the number of €0.50 increases.

(a) Show that the revenue (*opbrengst*) is $R = (10 + 0.5n)(300 - 10n)$.
(b) Which ticket price gives the maximum revenue? How many tickets are then sold, and what is the revenue?

**20. Cycling.** Sanne cycles $60$ km. If she had cycled $5$ km/h faster, she would have arrived $1$ hour earlier. Let $v$ be her speed in km/h.

(a) Explain why $\dfrac{60}{v} - \dfrac{60}{v+5} = 1$.
(b) Calculate her speed.

**21. Finding the formula.**

(a) The parabola $y = x^2 + bx + c$ passes through $(1, 0)$ and $(5, 0)$. Find $b$ and $c$ and the vertex.
(b) A parabola has vertex $(2, -1)$ and passes through $(0, 3)$. Find its formula in the form $y = ax^2 + bx + c$.

**22. Rectangle with conditions.** The length of a rectangle is $3$ cm more than its width $w$. The area must be at least $40$ cm$^2$ and at most $70$ cm$^2$. Which widths are possible?

## Challenge

**23.** (a) Show that $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$. Hint: write $x^4 + 4 = (x^2 + 2)^2 - 4x^2$.

(b) Prove that $n^4 + 4$ is a prime number for only one positive integer $n$.

**24.** Find all positive integers $n$ for which $n^2 + 60$ is a perfect square.

**25.** Solve $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.

**26.** (a) Prove that for all real numbers $a$, $b$ and $k$ the equation $(x - a)(x - b) = k^2$ has at least one real solution.

(b) Prove that for all positive numbers $p$ and $q$: $\dfrac{p+q}{2} \geq \sqrt{pq}$, with equality only when $p = q$. Use it to show that among all rectangles with perimeter $P$ the square has the largest area.

**27.** The roots of $x^2 - 5x + 3 = 0$ are $x_1$ and $x_2$. Without calculating the roots, find

(a) $x_1^2 + x_2^2$ (b) $\dfrac{1}{x_1} + \dfrac{1}{x_2}$ (c) $(x_1 - x_2)^2$ (d) $x_1^3 + x_2^3$

**28.** Find a quadratic equation $x^2 + bx + c = 0$ with integer coefficients whose roots are the squares of the roots of $x^2 - 4x + 1 = 0$.

**29.** A number $x$ satisfies $x + \dfrac{1}{x} = 3$.

(a) Without solving for $x$, compute $x^2 + \dfrac{1}{x^2}$ and $x^3 + \dfrac{1}{x^3}$.

(b) Now solve the equation and check one of your answers to (a).

**30.** (a) For which values of $m$ does $(m - 1)x^2 - 2mx + m + 2 = 0$ have exactly one solution? (Careful.)

(b) For which values of $m$ does the line $y = mx$ touch the parabola $y = x^2 + 1$? Give the points of contact.